

**SEMESTER 2
2022-2023**

**CS423
Designing for Virtual Environments**

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Time allowed: 2 hours

Answer at least **three** questions

Your mark will be based on your best **three** answers

All questions carry equal marks

Instructions

	Yes	No	N/A
Formulae and Tables book allowed (<i>i.e. available on request</i>)		X	
Formulae and Tables book required (<i>i.e. distributed prior to exam commencing</i>)		X	
Statistics Tables and Formulae allowed (<i>i.e. available on request</i>)		X	
Statistics Tables and Formulae required (<i>i.e. distributed prior to exam commencing</i>)		X	
Dictionary allowed (<i>supplied by the student</i>)		X	
Non-programmable calculator allowed		X	
Students required to write in and return the exam question paper		X	

- [25 marks]**
- 1** (a) Briefly discuss the concepts of presence and immersion in the context of VR and describe how they relate to each other. [12 marks]
- (b) What is interpupillary distance and why does it matter for VR? [8 marks]
- (c) What was the 'Sensorama' and what did it try to archive? [5 marks]
- [25 marks]**
- 2** (a) Briefly describe and highlight the advantages and drawbacks of the different forms of haptic displays in relation to VR. [10 marks]
- (c) Describe the difference between isomorphic and non-isomorphic manipulation in the context of VR with an example each. [15 marks]
- [25 marks]**
- 3** (a) One high level concept of good VR content is to focus on its core experience. Describe what this means and demonstrate it with an example that relates to VR. [10 marks]
- (b) What is Gestalt psychology? Further explain three principles with an example each that relate to VR. [15 marks]
- [25 marks]**
- 4** In the lecture we discussed Ivan Sutherland's paper "The Ultimate Display" that you have also been provided with to read on your own. [25 marks]
- Discuss the significance of the paper in relation to technology in general and to Virtual Reality in particular. Which aspects still matter and why? [Max 1 page]